

High performance concrete

Classification

Parameter	High Strength	Very High Strength	Ultra High Strength
Compressive Strength (N/mm ²)	42-100	100-150	> 150
w/(c+p) ratio or w/b ratio	0.45-0.30	0.30-0.24	< 0.24
Chemical admixture	WRA/HRWRA	HRWRA	HRWRA
Mineral admixture	Fly ash or combined with silica fume	Silica fume*	Silica fume*
Permeability coefficient (cm/s)	10 ⁻¹¹	10 ⁻¹²	< 10 ⁻¹⁴
Freeze-thaw protection	Air entrainment	Air entrainment	No freezable water

WRA: Water reducing admixture

*** May also contain fly ash**

HRWRA: High range water reducing admixture

High performance concrete

Requirements of making high performance concrete

- Low water-to-binder (w/b) ratio
- Smaller maximum size of aggregate (MSA)
- Fine aggregate with higher fineness modulus
- Use of mineral admixtures
- Use of chemical admixtures
- Binder-chemical admixture compatibility
- Proper curing

Mix proportioning of high performance concrete

- The mix proportioning method is a combination of empirical results and mathematical calculations based on absolute volume method.
- The water contributed by superplasticizer is considered as part of the mixing water.

Step 1: Selection of w/b ratio

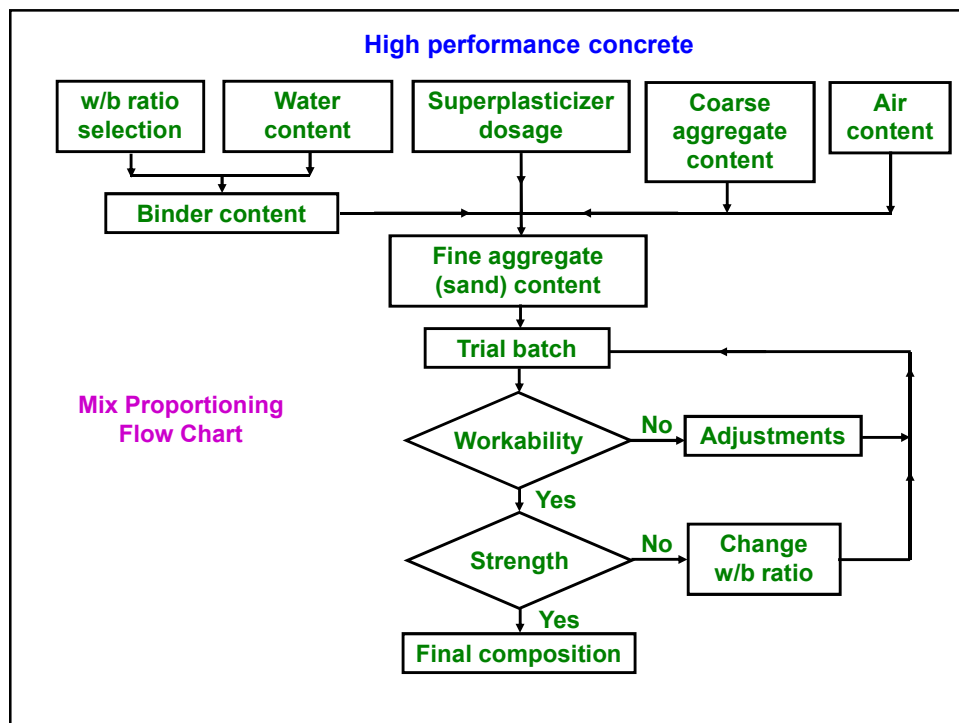
Step 2: Water content

Step 3: Superplasticizer dosage

Step 4: Coarse aggregate content

Step 5: Air content

Step 6: Fine aggregate content



Superplasticizer

M = mass of superplasticizer

V = volume of superplasticizer

G = specific gravity of superplasticizer

M_s = mass of solids in superplasticizer

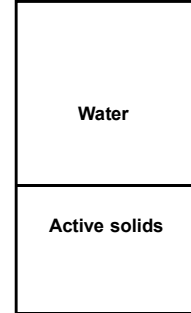
V_s = volume of solids in superplasticizer

M_w = mass of water in superplasticizer

V_w = volume of water in superplasticizer

s (%) = solids content of superplasticizer

$$M = M_w + M_s$$



Schematic representation of
a Superplasticizer

Superplasticizer

If **M** is in gram (g) and **V** is in cubic centimeter (cm³),

$$G = \frac{M}{V} \dots\dots\dots (1)$$

$$s (\%) = \frac{M_s}{M} \times 100 \dots\dots\dots (2)$$

$$M_s = \frac{G \times V \times s}{100} \dots\dots\dots (3)$$

$$M_w = G \times V \left[\frac{(100 - s)}{100} \right] \dots\dots\dots (4)$$

$$M_s = \frac{B \times d}{100} \dots\dots\dots (5)$$

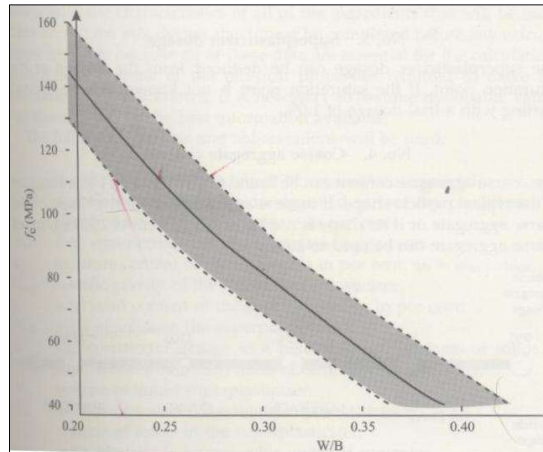
$$V = \frac{B \times d}{G \times s} \dots\dots\dots (6)$$

B = total binder content in concrete

d (%) = dosage of solids of superplasticizer at saturation point

Mix proportioning of high performance concrete

- **Water-binder ratio (w/b ratio):** It is obtained from the plot of w/b ratio vs. compressive strength .



w/b ratio vs. compressive strength

Mix proportioning of high performance concrete

- **Water content:** It is selected from the values corresponding to the saturation point of superplasticizer.

Saturation point (%)	0.6	0.8	1.0	1.2	1.4
Water content (l/m ³)	120-125	125-135	135-145	145-155	155-165

- To design a safe mix, **5 l/m³** of water can be added to the above values.

- if saturation point of the superplasticizer is not known, start with a water content of **145 l/m³**.

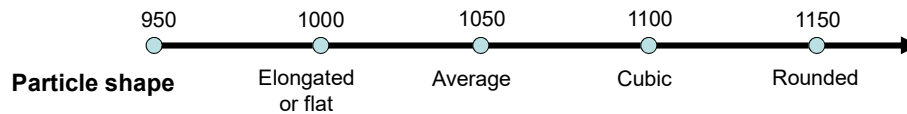
- **Superplasticizer dosage:** It can be calculated from the dosage at saturation point.

- If the saturation point is not known, start with a trial dosage of **1%**.

Mix proportioning of high performance concrete

- **Coarse aggregate content:** It can be obtained corresponding to particle shape.

Coarse aggregate
content (kg/m^3)



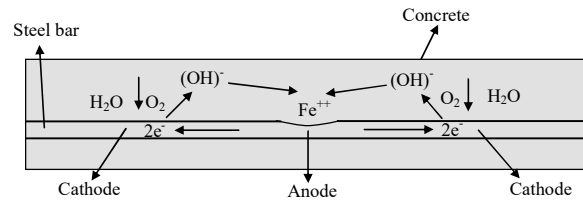
- Take coarse aggregate content of **1000 kg/m^3** , if there is any doubt about the shape of coarse aggregate or if the shape is not known.

- **Air content:** A value of **1.5%** can be taken as an initial estimate of entrapped air content.

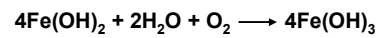
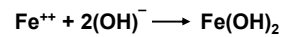
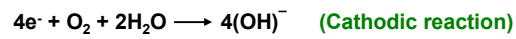
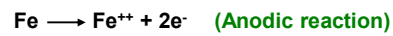
Corrosion of steel reinforcement in concrete

- ✓ Corrosion of steel reinforcement: an electrochemical process.
- ✓ In highly alkaline environment of concrete (pH of about 13), passive layer is formed on steel surface.
- ✓ However, breakdown of the passive film occurs and corrosion initiation takes place due to:
 - i) **Carbonation**, which reduces the pH value to a non-protection level.
 - ii) Presence of significant amount of **chloride ions** at steel reinforcement level.

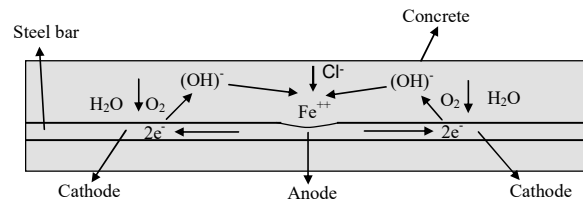
Corrosion of steel reinforcement in concrete



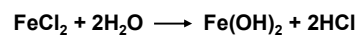
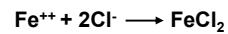
Schematic representation of rebar corrosion process



Corrosion of steel reinforcement in concrete



Schematic representation of corrosion process of steel reinforcement in the presence of chloride ions



Corrosion of steel reinforcement in concrete

- ✓ Chloride ions are considered to be the primary cause of steel reinforcement corrosion in concrete than that due to carbonation.
- ✓ **Generalized corrosion** takes place in presence of carbonation.
- ✓ **Pitting corrosion** takes place in presence of chloride ions.
- ✓ However, in the presence of large amount of chloride ions, generalized corrosion takes place.

Chloride induced corrosion

Sources of chloride in concrete

✓ Fresh concrete

- Mixing water
 - Chloride contaminated aggregates
 - Admixtures containing chloride
- } Internal chloride

✓ Hardened concrete

- Application of deicing salts
 - From sea water in marine structures
 - From soil and ground water containing chloride salts
- } External chloride

Chloride induced corrosion

Forms of chloride in concrete

✓ **Free chloride (Water soluble chloride)**

← Mainly responsible for corrosion of steel reinforcement

✓ **Bound chloride**

- Chemically bound with hydrated C_3A to form Friedel's salt (Calcium chloroaluminate)
- Physically bound: adsorbed on C-S-H.

- ✓ **Total chloride content:** sum of free chloride and bound chloride contents.

Critical chloride content:

- ✓ Chloride content at steel reinforcement level that results in corrosion initiation of steel reinforcement.

Chloride induced corrosion

Critical chloride content depends on:

- **Cement type**
- **C_3A content of cement**
- **Concrete mix proportion**
- **Steel type**
- **Source of chloride**

Carbonation induced corrosion

- ✓ Diffusion of CO_2 into concrete.
- ✓ CO_2 gets dissolved in concrete pore solution and results in formation of carbonic acid.
- ✓ Depletion in reserve level of calcium hydroxide in concrete.
- ✓ Thus, lowering the pH value to about 9.
- ✓ At this low pH value, passive layer is destroyed and corrosion initiation of steel reinforcement takes place.

Carbonation

Actual rate of carbonation depends on:

- Permeability and moisture content of concrete
- CO_2 content and relative humidity of ambient medium

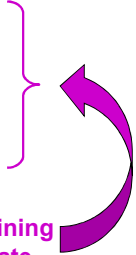
Extent of carbonation

- ✓ Extent of carbonation can be determined by treating a freshly broken concrete surface with Phenolphthalein.
 - Carbonated portion: uncoloured
 - Free $\text{Ca}(\text{OH})_2$: coloured pink

Corrosion monitoring techniques

- ✓ Electrochemical techniques are the most widely used techniques for the study of rebar corrosion.
- ✓ **Reason:** Increased number of structures require an assessment of their residual health condition due to premature deterioration caused by steel reinforcement corrosion.
- ✓ **Commonly used techniques:**
 - Half-cell potential measurement
 - Linear polarization resistance (LPR) technique
 - Electrochemical impedance spectroscopy (EIS) technique
 - Tafel plot technique
 - Gravimetric (mass loss) measurement
 - Resistivity measurement

For determining corrosion rate



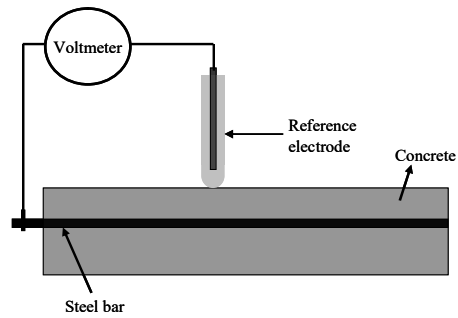
Half-cell potential measurement

- ✓ **Half-cell potential** is a measure of electrode potential that represents the potential difference between the steel and the adjacent concrete electrolyte.
- ✓ It gives an indication of the relative probability of occurrence of corrosion activity.
- ✓ The half-cell potential of steel reinforcement is measured with reference to a standard reference electrode.

For half-cell potential measurement:

- A high input impedance voltmeter
- Reference electrode (RE)
- Working electrode (WE): steel reinforcement

Half-cell potential measurement



ASTM C876 Criteria

Potential values

(Reference electrode Cu/CuSO₄)

> -200 mV

-200 mV to -350 mV

< -350 mV

Schematic representation of potential measurement on reinforced concrete specimen

Corrosion activity

Greater than 90% probability that no steel reinforcement corrosion is occurring.

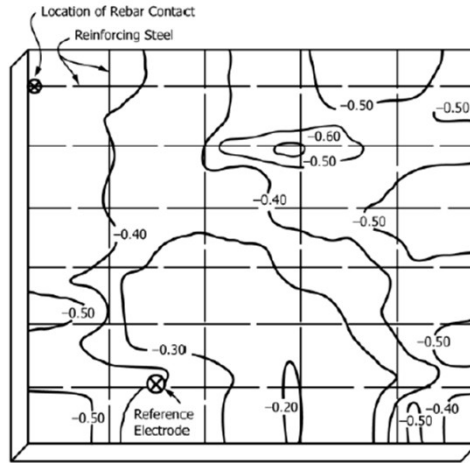
Corrosion activity is uncertain.

Greater than 90% probability of occurrence of steel reinforcement corrosion.

Types of reference electrode

- ✓ Standard hydrogen electrode (SHE)
- ✓ Copper-copper sulfate electrode (Cu/CuSO₄)
- ✓ Saturated calomel electrode (SCE)
- ✓ Silver-silver chloride electrode (Ag/AgCl)
- ✓ Mercury-mercurous sulfate electrode (Hg/HgSO₄)

Half-cell potential measurement



Equipotential contour map

Linear polarization resistance (LPR) technique

- ✓ **LPR technique:** a well-established method of determining the instantaneous corrosion rate of steel reinforcement in concrete.
- ✓ The steel reinforcement is polarized by the application of a small perturbation from the equilibrium potential through an **auxiliary electrode (AE)**.
- ✓ Steel reinforcement is polarized to $\pm$ (10 to 30 mV) from the equilibrium potential at a certain scan rate.
- ✓ Polarization resistance of steel (R_p) $R_p = \frac{\Delta E}{\Delta I}$ ΔE = change in potential
 ΔI = resulting current

Stern-Geary equation

$$I_{corr} = \frac{B}{R_p}$$

Where I_{corr} = corrosion current density

B = Stern-Geary constant

$$B = (\beta_a \times \beta_c) / 2.3 (\beta_a + \beta_c)$$

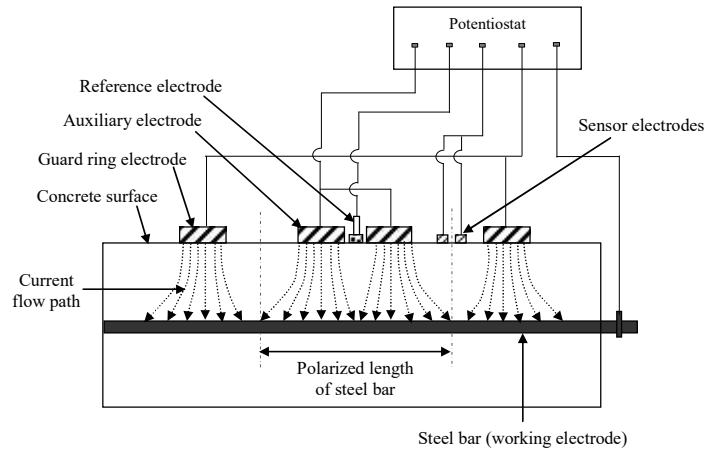
β_a = anodic Tafel constant

β_c = cathodic Tafel constant

B = 26 mV for steel in active condition
and

B = 52 mV for steel in passive condition

Linear polarization resistance (LPR) technique



Schematic representation of LPR measurement with guard ring arrangement

Gravimetric (mass loss) measurement for corrosion rate

For steel reinforcement

✓ Cleaning procedure as specified in **ASTM G1 – 03**

✓ **Cleaning solution:**

- 500 ml of Hydrochloric acid
- 3.5 g of Hexamethylene tetramine
- Reagent water to make 1000 ml

$$\text{Corrosion rate} = \frac{K \times W}{A \times T \times D}$$

K = a constant

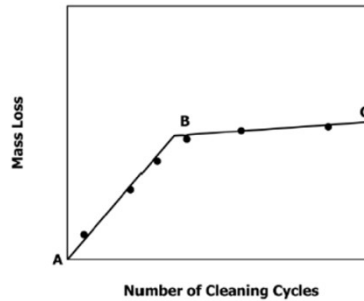
W = mass loss in grams

A = corroded surface area of steel reinforcement in cm²

T = time of exposure in hours

D = density of steel in g/cm³

Gravimetric (mass loss) measurement for corrosion rate



Mass loss of corroded specimen resulting from repetitive cleaning cycles

Resistivity measurement

- ✓ Electrical resistivity of concrete (Ohm-m).
- ✓ Resistivity of concrete is influenced by w/c ratio, moisture content, composition of concrete, temperature etc.

Four electrodes method (or Wenner four probe method)

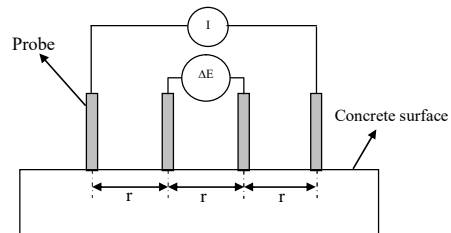
ρ = resistivity of concrete

$$\rho = 2\pi r \frac{\Delta E}{I}$$

Where ΔE = potential drop across the inner electrodes

I = current flowing between the outer electrodes

r = electrode spacing



Resistivity measurement of concrete using four electrodes method

Preventive and protection measures for decreasing corrosion of steel reinforcement

Preventive measures

- Low w/c ratio
- Suitable type of cement
- Mineral admixtures (fly ash, silica fume, slag etc.)
- Corrosion resistant steel reinforcement

Protection measures

- Corrosion inhibitors
- Cathodic protection
- Different types of surface treatment for concrete

References

- Nawy, E. G. Fundamentals of High-Performance Concrete, John Wiley & Sons, Inc., Second edition, 2001.
- Aitcin, P. C. High-Performance Concrete, E & FN SPON, 1998.
- Neville, A. M., and Brooks, J. J. Concrete Technology, Pearson Education, Fourth Indian reprint, 2004.
- Pradhan, B., and Bhattacharjee, B. Performance evaluation of rebar in chloride contaminated concrete by corrosion rate, Construction and Building Materials, Vol. 23, 2346-2356, 2009.
- Pradhan, B., and Bhattacharjee, B. Rebar corrosion in chloride environment, Construction and Building Materials, Vol. 25, 2565-2575, 2011.
- Pradhan, B. Monitoring techniques and preventive measures for rebar corrosion in concrete - a review, International Journal of 3R's, Vol. 1, 33-41, 2010.
- Bertolini, L., Elsener, B., Pedferri, P., and Polder, R. Corrosion of Steel in Concrete, Prevention, Diagnosis, Repair, WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim, 2004.
- Schiessl, P. Corrosion of Steel in Concrete, Report of the Technical Committee 60-CSC RILEM, Chapman and Hall, 1988.

References

- ASTM C 876 - 15. Standard test method for corrosion potentials of uncoated reinforcing steel in concrete, West Conshohocken, PA, 2015.
- Dehwah, H. A. F., Maslehuddin, M., and Austin, S. A. Long-term effect of sulfate ions and associated cation type on chloride-induced reinforcement corrosion in Portland cement concretes, *Cement & Concrete Composites*, Vol. 24, 17-25, 2002.
- Pradhan, B. Corrosion behavior of steel reinforcement in concrete exposed to composite chloride-sulfate environment, *Construction and Building Materials*, Vol. 72, 398-410, 2014.
- ASTM G 1 – 03 (Reapproved 2011). Standard practice for preparing, cleaning, and evaluating corrosion test specimens, West Conshohocken, PA, 2011.